



CPCS351 Syllabus

Faculty of Computing and Information Technology
Department of Computer Science



Instructor

Course Catalogue

Phone

Course Number: CPCS351

Email

Course Title: Software Engineering 1

Course Credits: 3

Prerequisite: CPCS204 (Data Structure)

Office Location

Course Coordinators: Dr. Asif Irshad Khan, aikhan@kau.edu.sa

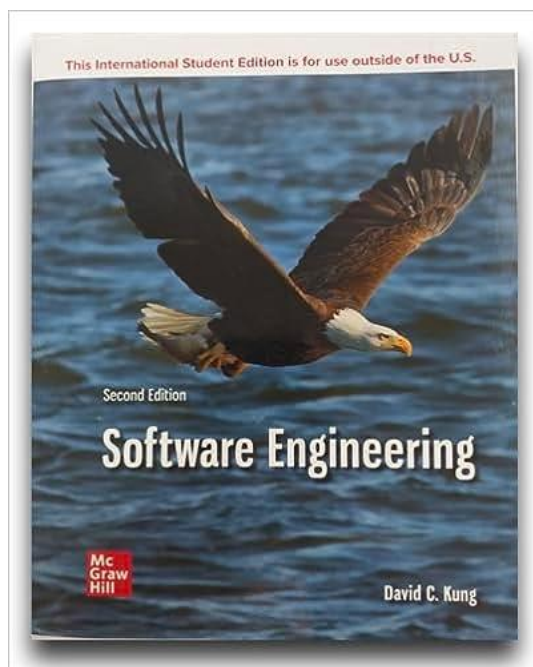
Office Hours

Course Description and Objective

Teaching Schedule

The objective of this course to introduce students to the basic concepts of software engineering as they relate to the development of medium to large scale software systems. Topics include the software life cycle, requirements development, object orientation, software architecture & design, and software testing. Students are expected to learn how to apply such principles to a real world problem. A term project of a medium size is required.

Textbook



Software Engineering: An Agile Unified Methodology 2nd Edition (ISE), by David C. Kung (Author), Publisher : McGraw Hill; 2nd Edition (ISE) (January 1, 2023), ISBN-13 : 978-1265242435

Course Learning Outcomes (CLO) and Student Outcomes (SO)

1. Explain the concept of a software lifecycle and provide an example, illustrating its phases including the deliverables that are produced. (1)
2. Define software quality and the role of quality assurance activities in the software process. (1)
3. Differentiate among the phases of software development (1)
4. Explain the different software development models (traditional, iterative, incremental and agile) and their processes correctly. (1)
5. Justify both functional and non-functional requirements in a given requirements specification for a software system (2)
6. Apply key elements and common methods for elicitation and analysis to produce a set of software requirements for a medium-sized software system. (2)
7. Articulate design principles including separation of concerns, information hiding, coupling and cohesion, and encapsulation. (2)
8. Create appropriate design models for the structure and behavior of software products from their requirements specifications. (2)
9. Learn UML Use Case diagram notation and style to depict high-level user requirements and be able to write up Use Case Centric Functional Requirements Specifications in a typical commercial system. (2)
10. Depict UML diagrams that illustrate the structure and the dynamic behavior of a system (Class diagrams, sequence diagrams, activity diagrams and state diagrams). (2)
11. Identify popular software architectures (such as 3-tier, pipe-and-filter, and client server) and the influence of an architecture on the design of a software system. (2)
12. Apply design patterns to enhance class diagram in the design model (2)
13. Communicate orally and in writing the results of applying the acquired knowledge of software engineering phases of analysis and design and their relevant techniques to a small sized group project. (3)
14. Describe and distinguish among the different types and levels of testing (unit, integration, systems, and acceptance). (1)

Grade Distribution

Assessment	Percentage
Group Project	30
Research Component	10
Lab Graded Activities	10
Midterm Exam	20
Comprehensive Final Exam	30

Deliverables' Assignments and Important Dates

Assessment	Assigned Date
Project (Phases I)	W3
Project (Phases II)	W7
Project (Phases III)	W10
Project (Phases IV)	W14
Research Component	W10
Lab Graded Activity #1	W4
Lab Graded Activity #2	W7
Lab Graded Activity #3	W10
Lab Graded Activity #4	W12
Midterm Exam	W8: TBC
Comprehensive Final Exam	W16/ W17 / W8: TBC

Course Schedule

Week #	Chapter	Lecture Topics	Lab Topics	Sections covered	What is Due?
1	Chapter 1	Introduction to the course (a) What is Software Engineering (SE)? (b) Why SE? (c) Software Life Cycle Activities		1.1, 1.2, 1.3	1- Buy Book 2- CLO #1
2	Chapter 1,19	What is Software Quality Assurance? Verification Vs. Validation	Introduction	1.3.2, 19.2	Determining groups and project topics
3	Chapter 19, Chapter 3	What are the most important attributes of software quality?	Software Designing Tools , Visual	19.2, 3.3, 3.4, 3.5,3.6	2. CLO # 2 CLO #3 Project – Phase 1

		Short Term Vs. Long Term Quality System Req. Def. System Arch. Design Subsystems Development System Integration Testing and Deployment.	Paradigm or Star UML.		
4	Chapter 2	Process and Methodology Water Fall Prototyping Evolutionary Spiral Agile	Reviewing the Software Development Models	2.3, 2.5.1,2.5.2,2.5.3,2.5.7	CLO #4
5	Chapter 4, 7	What is Requirements Elicitation? Importance of Requirements Elicitation. Challenges of Requirements Elicitation. Types of Requirement. Steps of Requirements Elicitation. What is an Actor?	The concept of system requirements, and help them differentiate between functional and nonfunctional requirements	4.1, 4.2, 4.3, 4.4, 4.5,	
6	Chapter 4, 7	What is a Use Case? Business Process, Operation, And Action. Steps for deriving Use cases from Requirements Guidelines for Use Case Derivation.	UML Use case Modeling	7.1, 7.2, 7.3, 7.4, 7.5,7.6	1. CLO # 5,6, 9 2 - Project – Phase 2 4 – Research Component
7	Chapter 5	collecting information about the application domain;	UML Domain Modeling	5.1,5.2,5.3, 5.7	

		brainstorming; classifying brainstorming results; visualizing the domain model using a UML class diagram; and performing inspection and review			
8	Chapter 6	Articulate design principles including separation of concerns, information hiding, coupling and cohesion, and encapsulation	Graded Lab Work1	6.5 (all)	CLO # 7 - Graded Lab Work1
9	Chapter 9	Object interaction modeling helps the development team understand the existing business processes, and design object interaction behaviors to improve the business. Actor-system interaction modeling and object interaction modeling deal with foreground processing and background processing of a use case, respectively.	UML Sequence Modeling	9.1,9.2	
10	Chapter 14	What is Object State Modeling? Why Object State Modeling? Commonly used UML state	UML State and Activity Modeling	13.1,13.2,13.4.5,13.4.6 14.4, 14.5, 14.5.2, 14.5.3	1. CLO# 8, 10 2. Project Presentation 3. Project – Phase 3

		diagrams notations. Design State Diagram. Activity modeling, branching, concurrency, synchronous and asynchronous behavior. modeling and design of transformational systems.			4. Graded Lab Work2
11	Chapter 6	What is architecture design? Importance of architecture design. architecture design Process architecture Style and Package Diagram	Graded Lab Work3	6.1 to 6.4	1. CLO# 11 3. Graded Lab Work3
12	Chapter 10	What is Design Patters? Why Design Patters? The Controller Patter. The Expert Pattern. The Creator Pattern	Design Pattern and Architecture design	All Sections	1. CLO # 12 2. Exam2 3. Project – Phase 4 4. Graded Lab Work4
13	Chapter 20	What is Software Testing? Why Software Testing? Types of software testing	Junit Testing	20.1, 20.2	CLO # 14
14	Chapter 20	Test Design techniques		20.1, 20.2	Final Project submission and Presentation
15	Course revision	Finals			

Course General Policies

- Students are encouraged to ask and discuss the materials during lectures, labs, and office hours. **If students require further assistance in group projects, they should contact instructor or the lab instructor for help.** The goal of the lab instructor is to provide the facilitation necessary for students help in design and analysis of the course project they chosen.
- **Students are responsible for checking their email or Blackboard regularly for important announcements, assignments posting, and material updates.** Blackboard will be an essential element of the course, and It is the students' responsibility to check Blackboard before class for any updates.
- There will be **NO make-up exams**. Students must submit an official medical report to the academic affairs, within 5 days after the exam she was absent from. The Academic affairs must check the validity of the medical report, then send it to the concerned instructor. On that case, the marks will be considered from the average of the First/or the second and the final exam.
- **Class attendance** is mandatory and will be taken immediately at the beginning of each class. If you miss more than 25% of the lectures, you will receive a DN.

Assessment tool policies Labs:

- One of the primary purposes of the lab is to reinforce and practice the information presented in the in-class lectures.
- During lab time, lab instructors will provide examples and assistance with concepts covered in the book and during lectures, and occasionally will introduce new material as well.
- Lab time will be your opportunity to:
 - Become accustomed to the CS department computing environment
 - Learn how to use the Modeling tools.
 - Gain an experience and ask questions about design and analysis of the software system.
- **Students during the labs must work individually and questions should be directed to the lab instructors only.**
- Students should come fully prepare for lab by attending lectures and completing any assigned reading, as well as carefully reviewing and, in appropriate cases, getting started on the lab assignments provided.
- There should be no lectures given during the lab sessions, the most duration of the lab should be utilized by the student to implement the given problems.
- Students are expected to submit solutions to the lab instructor in order to assign the mark activities for students in each lab. Therefore, **lab attendance AND participation are mandatory.**
- **Students must submit at least 70% of the problem solution in the lab to get the participation mark.** If the student did not finish the lab requirement, the instructor could allow the student to complete the problem at home in the same day of the lab (Note that this chance should be given once to the student during the term).
- **Students must attend at least 75% of the labs. If they miss more than 25% of the labs, they will not be allowed to take the lab final exam.**

Plagiarism Policies

- Students are allowed to discuss concepts/ideas related to assignments with other students, but students are not allowed to copy a segment of other students' analysis and design copy and modify another student's work, work with another student on any part of online resources.
- The submitted assignments will be checked via a **plagiarism system** to detect the similarity between each of the submitted assignments as well as online resources. **This procedure will be applied to all submitted project reports in all the sections (including both the boys and girls campuses) .**